Research Methodology in CS

Concepts, methods and reflections

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Roadmap

1. Defining a research problem
2. Reasoning and hypotheses
3. Structure of a research paper
4. Experiments and evaluation
5. Data mining and KDD
6. Publishing, ethics and IP
7. My reflections

1. Defining a Research Problem

Key terms

- Research problem
- Research question
- Scope and objectives

Ask yourself

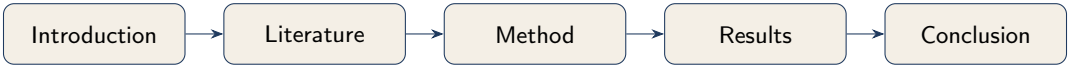
Is it significant? Is it investigable? Is it measurable?

2. Reasoning and Hypotheses

Type	Direction	Your example
Deductive	general $\rightarrow$ specific	[edit]
Inductive	specific $\rightarrow$ general	[edit]
Abductive	observation $\rightarrow$ best explanation	[edit]

Hypothesis testing: null hypothesis H_0, alternative H_1, significance level α, p-value.

3. Structure of a Research Paper



- Supporting parts: references, data and code availability, acknowledgements, funding
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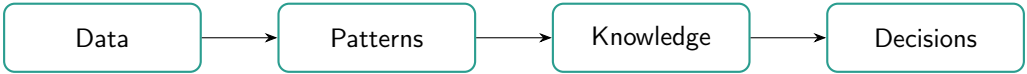
4. Experiments and Evaluation

- Train / validation / test split
- Baselines and comparison
- Overfitting and outliers
- Statistical significance

Metrics

Accuracy, precision, recall, F1, ROC–AUC, confusion matrix

5. Data Mining and KDD



- Clustering: k-medoids, hierarchical
- OLTP vs OLAP, data warehouse
- OLAP operations and ROLAP / MOLAP / HOLAP
- Neuro-fuzzy: Mamdani vs Sugeno

6. Publishing, Ethics and IP

- Research design: pilot study, case study, interviews, projective techniques
- Bloom's taxonomy
- Writing: originality, paraphrasing, plagiarism, AI-use disclosure
- Submission, review and revision
- Intellectual property and patents

Thank you